Systematic Search for Singularities in 3D Euler-Voigt Flows

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Funded by NSERC (Canada), Computing Time Provided by COMPUTE CANADA

March 9, 2026

Agenda

Singularities & Euler-Voigt Flows

- Navier-Stokes system ($\Omega = [0, L]^3$)

$$\begin{cases} \partial_t \mathbf{u} + (\mathbf{u}\cdot\boldsymbol{\nabla})\mathbf{u} + \boldsymbol{\nabla} p - \nu\Delta\mathbf{u} = \mathbf{0}, & \text{in } \Omega \times (0, T] \\ \boldsymbol{\nabla}\cdot\mathbf{u} = 0, & \text{in } \Omega \times (0, T] \\ \mathbf{u} = \boldsymbol{\eta} & \text{in } \Omega \text{ at } t = 0 \\ \text{Periodic Boundary Conditions} & \text{on } \partial\Omega \times (0, T] \end{cases}$$

- **The Big Question:**

  Given a smooth initial condition $\boldsymbol{\eta}$, does the Navier-Stokes (and Euler) system always admit smooth solutions $\mathbf{u}(t)$ for arbitrarily long times t?

- One of the Clay Institute “Millennium Problems” ($ 1M prize!)

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- $\alpha > 0$ is the regularization parameter with units of length. Acts as a 'spectral filter'. The Euler-Voigt system is globally well-posed given smooth initial conditions [CITATION].
- As $\alpha \to 0$ the behaviour can be used to determine if the Euler equations will develop a finite-time singularity.

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- Proven by Theorem 1 in [COMPUTATIONAL INVESTIGATION CITATION]. WHY DONT WE JUST GO DIRECTLY TO THEOREM 5.2 CONCLUSION

Theorem

Let $\boldsymbol{\eta}_\alpha \in H^s$, $s \geq 3$ with $\boldsymbol{\nabla} \cdot \boldsymbol{\eta}_\alpha = 0$ and let $\boldsymbol{u}_\alpha$ be the corresponding unique solution of Euler-Voigt. Suppose that there exists a time $T^* > 0$ such that

$$\limsup_{\alpha \to 0^+} \left(\alpha \sup_{t \in [0, T^*]} \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2} \right) > 0. \tag{1}$$

Then the Euler equations with the initial condition $\boldsymbol{\eta}_\alpha$ must develop a singularity within the interval $[0, T^*]$.

- Theorem 5.1 of [2nd COMPUTATIONAL INVESTIGATION CITATION]

Theorem

Given initial data $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\mathbb{T}^3)$ for some $s \geq 3$, let

$$\mathbf{u}, \boldsymbol{u}_\alpha \in C([0,T], H^s(\mathbb{T}^3)) \cap C^1([0,T], H^{s-1}(\mathbb{T}^3))$$

be the corresponding solutions to Euler and Euler-Voigt, respectively, where $0 < T < T_{max}$ and T_{max} is the maximal time for which a solution to Euler exists and is unique. For all $t \in [0,T]$, we have

$$\|\mathbf{u}(t) - \boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\mathbf{u}(t) - \boldsymbol{u}_\alpha(t))\|_{L^2}^2 \leq$$
$$\left(\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2\right) e^{Ct} + C\alpha^2 (e^{Ct} - 1).$$

- Modified from Theorem 5.2 [2nd COMPUTATIONAL INVESTIGATION CITATION]

Theorem

Given initial data $\boldsymbol{\eta}, \boldsymbol{\eta}_\alpha \in H^s(\Omega)$ *for some* $s \geq 3$ *satisfying*

$$\|\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2 \|\boldsymbol{\nabla}(\boldsymbol{\eta} - \boldsymbol{\eta}_\alpha)\|_{L^2}^2 \to 0, \quad \textit{as } \alpha \to 0,$$

let $\mathbf{u}, \boldsymbol{u}_\alpha \in C([0,T], H^s(\Omega)) \cap C^1([0,T], H^{s-1}(\Omega))$ *be the corresponding solutions to Euler and Euler-Voigt, respectively. If there exists a finite time* $T^* > 0$ *such that the solutions* $\boldsymbol{u}_\alpha$ *satisfies*

$$\limsup_{\alpha \to 0} \left(\alpha^2 \sup_{t \in [0, T^*]} \|\boldsymbol{\nabla} \boldsymbol{u}_\alpha(t)\|_{L^2}^2 \right) > 0,$$

then $\mathbf{u}$ *develops a singularity in the interval* $[0, T^*]$.

Proof.

Suppose that the solution **u** of the Euler equations Euler with the initial condition $\boldsymbol{\eta}$ is regular on $[0, T^*]$. Since the Euler-Voigt equations are globally well-posed, we have

$$\|\boldsymbol{u}_\alpha(t)\|_{L^2}^2 + \alpha^2\|\nabla\boldsymbol{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2\|\nabla\boldsymbol{\eta}_\alpha\|_{L^2}^2,$$

thus

$$\alpha^2\|\nabla\boldsymbol{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2\|\nabla\boldsymbol{\eta}_\alpha\|_{L^2}^2 - \|\boldsymbol{u}_\alpha(t)\|_{L^2}^2.$$

Taking the supremum over $t \in [0, T^*]$ of the above equation, we obtain that

$$\sup_{t\in[0,T^*]} \alpha^2\|\nabla\boldsymbol{u}_\alpha(t)\|_{L^2}^2 = \|\boldsymbol{\eta}_\alpha\|_{L^2}^2 + \alpha^2\|\nabla(\boldsymbol{\eta}_\alpha)\|_{L^2}^2 - \inf_{t\in[0,T^*]} \|\boldsymbol{u}_\alpha(t)\|_{L^2}^2.$$

Proof (Cont.)

Taking the limsup as $\alpha \to 0$, we obtain that

$$\limsup_{\alpha\to 0}\left(\sup_{t\in[0,T^*]}\alpha^2\|\nabla \boldsymbol{u}_\alpha(t)\|_{L^2}^2\right) =$$
$$\limsup_{\alpha\to 0}\left(\|\boldsymbol{\eta}_\alpha\|_{L^2}^2+\alpha^2\|\nabla(\boldsymbol{\eta}_\alpha)\|_{L^2}^2-\inf_{t\in[0,T^*]}\|\boldsymbol{u}_\alpha(t)\|_{L^2}^2\right).$$

We now make use of Theorem (1.2), which demonstrates that the convergence of the solutions of the Euler-Voigt equations to the solution of the Euler equations is uniform on $[0, T^*]$. Therefore we can reduce the above equation to

$$\limsup_{\alpha\to 0}\left(\sup_{t\in[0,T^*]}\alpha^2\|\nabla \boldsymbol{u}_\alpha(t)\|_{L^2}^2\right) = \|\boldsymbol{\eta}\|_{L^2}^2-\|\mathbf{u}(t)\|_{L^2}^2.$$

Proof (Cont.)

Since the energy of regular solutions of the Euler equations is conserved, we have

$$\limsup_{\alpha\to 0}\left(\sup_{t\in[0,T^*]}\alpha^2\|\nabla \boldsymbol{u}_\alpha(t)\|_{L^2}^2\right)=0,$$

which contradicts the hypothesis in the theorem that the above expression is positive. The second equality holds because

$$\begin{aligned}\limsup_{\alpha\to 0}\left|\inf_{t\in[0,T^*]}\|\boldsymbol{u}_\alpha(t)\|_{L^2}-\|\boldsymbol{\eta}\|_{L^2}\right| &\le \limsup_{\alpha\to 0}\|\boldsymbol{u}_\alpha(t_\alpha)-\boldsymbol{\eta}\|_{L^2}^2\\ &\le \limsup_{\alpha\to 0}\|\boldsymbol{u}_\alpha(t_\alpha)-\mathbf{u}(t_\alpha)\|_{L^2}^2+\|\mathbf{u}(t_\alpha)-\boldsymbol{\eta}\|_{L^2}^2\\ &= 0.\end{aligned}$$

□

- We consider analytic initial conditions that belong to the following Gevrey class

$$G^\sigma := \left\{ \mathbf{v} \in H^s(\Omega) : \|\mathbf{v}\|_{G^\sigma} = \langle \mathbf{v}, \mathbf{v} \rangle_{G^\sigma}^{1/2} < \infty \right\}, \quad s \geq 3, \quad \sigma > 0.$$

- We select the following inner product

$$\forall \mathbf{v}, \mathbf{u} \in G^\sigma, \quad \langle \mathbf{v}, \mathbf{u} \rangle_{G^\sigma} := \sum_{\mathbf{j} \in \mathbb{Z}^3} (1 + |2\pi \mathbf{j}|^2)^s e^{4\pi\sigma|\mathbf{j}|} \hat{\mathbf{v}}_{\mathbf{j}} \cdot \overline{\hat{\mathbf{u}}_{\mathbf{j}}}$$
$$= \int_\Omega (1 + |D|^2)^s e^{2\sigma|D|} \mathbf{v} \cdot \mathbf{u} \, d\mathbf{x}.$$

- The operators $|D|$ and $e^{\sigma|D|}$ are defined via

$$\left[\widehat{|D| \boldsymbol{v}} \right]_{\boldsymbol{j}} := 2\pi |\boldsymbol{j}| \hat{\boldsymbol{v}}_{\boldsymbol{j}}, \qquad \left[\widehat{e^{\sigma|D|} \boldsymbol{v}} \right]_{\boldsymbol{j}} := e^{2\pi|\boldsymbol{j}|} \hat{\boldsymbol{v}}_{\boldsymbol{j}}.$$

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$$V := \{\boldsymbol{v} \in G^{\sigma} : \boldsymbol{\nabla} \cdot \boldsymbol{v} = 0, \quad \int_{\Omega} \boldsymbol{v} \, d\mathbf{x} = 0\}.$$

- We use a closed manifold $\mathcal{M}_1 \subset V$ defined as

$$\mathcal{M}_1 = \{\mathbf{v} \in V : \|\mathbf{v}\|_{\dot{H}^1} = C\}.$$

- Euler and Euler-Voigt time scaling property

$$\tilde{\mathbf{u}}(\mathbf{x}, t) = \lambda \mathbf{u}(\mathbf{x}, \lambda t), \qquad (6)$$

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- Taylor Green Vortex ($\Omega = [0, L]^3$)

$$\boldsymbol{\eta}_{TGV} = \begin{bmatrix} V_0 \sin(2\pi x/L)\cos(2\pi y/L)\cos(2\pi z/L) \\ -V_0 \cos(2\pi x/L)\sin(2\pi y/L)\cos(2\pi z/L) \\ 0 \end{bmatrix}$$

- Characteristic time-scale

$$t_c = \frac{L}{V_0}$$

- A [CITATION] showed evidence for finite-time singularity in Euler from η_{TGV} at $T^* \approx 4$. Used ($\Omega = [0, 2\pi]^3$) and $V_0 = 1$. We use $\Omega = [0, 1]^3$ and so scale down to $V_0 = 1/2\pi$.

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- Calculate C analytically

$$C = \|\boldsymbol{\eta}\|_{\dot{H}^1} = \|\nabla \boldsymbol{\eta}\|_{L^2} = \left[\sum_{i,j=1}^{3} \int_{\Omega} (\partial_i \eta_j)^2 \, dx \right]^{1/2}$$

- Calculating for scaled η_{TGV}

$$\|\eta_{TGV}\|_{\dot{H}^1} = \sqrt{3}/2$$

- Create the manifold

$$\mathcal{M}_1 = \{\mathbf{v} \in V : \|\mathbf{v}\|_{\dot{H}^1} = \sqrt{3}/2\}.$$

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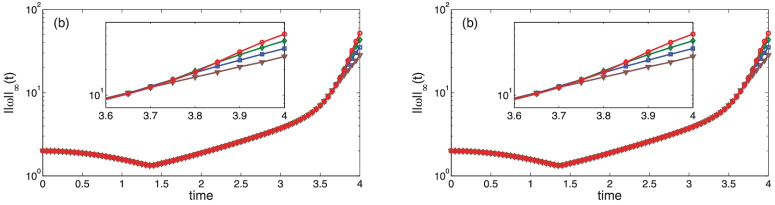
- Verify $\|\mathbf{v}\|_{\dot{H}^1}$ with vorticity

$$\|\omega\|_{L^\infty} = \|\boldsymbol{\nabla} \times \mathbf{u}\|_{L^\infty}$$



Optimization Problem for Euler-Voigt

- Define our objective function

$$\Phi_{\alpha,T}(\boldsymbol{\eta}) := \|\nabla \boldsymbol{u}_\alpha(T;\boldsymbol{\eta}_\alpha)\|_{L^2}^2 = \|\boldsymbol{u}_\alpha(T;\boldsymbol{\eta}_\alpha)\|_{\dot{H}^1}^2$$

- Given $\alpha, T \in \mathbb{R}^+$, find

$$\tilde{\boldsymbol{\eta}}_{\alpha,T} = \underset{\boldsymbol{\eta}_\alpha \in \mathcal{M}_1}{\operatorname{argmax}} \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha).$$

- If $\alpha^2 \Phi_{\alpha,T}(\tilde{\boldsymbol{\eta}}_{\alpha,T}) > 0$ as $\alpha \to 0$ then a singularity forms in Euler.

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- The first order finite difference equation for the Gâteaux differential is,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}_\alpha; \boldsymbol{\eta}'_\alpha) := \lim_{\epsilon \to 0} \frac{1}{\epsilon} \left[\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha + \epsilon \boldsymbol{\eta}'_\alpha) - \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha) \right].$$

- Perturbations

$$\boldsymbol{u}_\alpha = \boldsymbol{u}_\alpha + \boldsymbol{u}'_\alpha, \eta_\alpha = \eta_\alpha + \eta'_\alpha, \text{ and } p = p + p'$$

- Linearization of Euler-Voigt

$$\mathcal{L} \begin{bmatrix} \boldsymbol{u}'_\alpha \\ p' \end{bmatrix} := \begin{bmatrix} \partial_t \boldsymbol{u}'_\alpha + (I - \alpha^2 \Delta)^{-1} (\boldsymbol{u}_\alpha \cdot \nabla \boldsymbol{u}'_\alpha + \boldsymbol{u}'_\alpha \cdot \nabla \boldsymbol{u}_\alpha + \nabla p') = 0 \\ \nabla \cdot \boldsymbol{u}'_\alpha \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\boldsymbol{u}'_\alpha(x, 0) = \boldsymbol{\eta}'_\alpha$$

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- Integration by parts on the following duality-pairing relation

$$\begin{aligned}\left(\mathcal{L}\begin{bmatrix}\boldsymbol{u}'_\alpha\\ p'\end{bmatrix},\begin{bmatrix}\boldsymbol{u}^*_\alpha\\ p^*\end{bmatrix}\right) &:= \int_0^T\int_{\mathbb{T}^3}\mathcal{L}\begin{bmatrix}\boldsymbol{u}'_\alpha\\ p'\end{bmatrix},\begin{bmatrix}\boldsymbol{u}^*_\alpha\\ p^*\end{bmatrix}\,d\mathbf{x}\,dt\\ &=\left(\begin{bmatrix}\boldsymbol{u}'_\alpha\\ p'\end{bmatrix},\mathcal{L}^*\begin{bmatrix}\boldsymbol{u}^*_\alpha\\ p^*\end{bmatrix}\right)+\int_{\mathbb{T}^3}\boldsymbol{u}'_\alpha(\mathbf{x},T)\cdot\boldsymbol{u}^*_\alpha(\mathbf{x},T)\,d\mathbf{x}\\ &\quad-\int_{\mathbb{T}^3}\eta'(\mathbf{x})\cdot\boldsymbol{u}^*_\alpha(\mathbf{x},0)\,d\mathbf{x}\\ &=0.\end{aligned}$$

- Adjoint System

$$\mathcal{L}^*\begin{bmatrix}\boldsymbol{u}^*_\alpha\\ p^*_\alpha\end{bmatrix}:=\begin{bmatrix}-\partial_t\boldsymbol{u}^*_\alpha-(\nabla(I-\alpha^2)^{-1}\boldsymbol{u}^*_\alpha+(\nabla(I-\alpha^2)^{-1}\boldsymbol{u}^*_\alpha)^\intercal)\boldsymbol{u}_\alpha-\nabla p^*_\alpha\\ -\nabla\cdot\boldsymbol{u}^*_\alpha\end{bmatrix}=\begin{bmatrix}0\\ 0\end{bmatrix}$$

$$\boldsymbol{u}^*_\alpha(\mathbf{x},T)=2|D|^2\boldsymbol{u}_\alpha(\mathbf{x},T)$$

- Integration by parts on the following duality-pairing relation

$$\begin{aligned}\left(\mathcal{L}\begin{bmatrix}\boldsymbol{u}'_\alpha\\ p'\end{bmatrix}, \begin{bmatrix}\boldsymbol{u}^*_\alpha\\ p^*\end{bmatrix}\right) &:= \int_0^T\int_{\mathbb{T}^3}\mathcal{L}\begin{bmatrix}\boldsymbol{u}'_\alpha\\ p'\end{bmatrix}, \begin{bmatrix}\boldsymbol{u}^*_\alpha\\ p^*\end{bmatrix}\, d\mathbf{x}\, dt\\ &= \left(\begin{bmatrix}\boldsymbol{u}'_\alpha\\ p'\end{bmatrix}, \mathcal{L}^*\begin{bmatrix}\boldsymbol{u}^*_\alpha\\ p^*\end{bmatrix}\right) + \int_{\mathbb{T}^3}\boldsymbol{u}'_\alpha(\mathbf{x},T)\cdot\boldsymbol{u}^*_\alpha(\mathbf{x},T)\, d\mathbf{x}\\ &\quad - \int_{\mathbb{T}^3}\eta'(\mathbf{x})\cdot\boldsymbol{u}^*_\alpha(\mathbf{x},0)\, d\mathbf{x}\\ &= 0.\end{aligned}$$

- Adjoint System

$$\mathcal{L}^*\begin{bmatrix}\boldsymbol{u}^*_\alpha\\ p^*_\alpha\end{bmatrix} := \begin{bmatrix}-\partial_t\boldsymbol{u}^*_\alpha - (\boldsymbol{\nabla}(I-\alpha^2)^{-1}\boldsymbol{u}^*_\alpha + (\boldsymbol{\nabla}(I-\alpha^2)^{-1}\boldsymbol{u}^*_\alpha)^\intercal)\boldsymbol{u}_\alpha - \boldsymbol{\nabla} p^*_\alpha\\ -\boldsymbol{\nabla}\cdot\boldsymbol{u}^*_\alpha\end{bmatrix} = \begin{bmatrix}0\\ 0\end{bmatrix}$$

$$\boldsymbol{u}^*_\alpha(\mathbf{x},T) = 2|D|^2\boldsymbol{u}_\alpha(\mathbf{x},T)$$

- Substituting $\Phi_{\alpha,T}$ into the Gâteaux differential,

$$\Phi'_{\alpha,T}(\boldsymbol{\eta};\boldsymbol{\eta}') = 2\left\langle \boldsymbol{u}_\alpha(\mathbf{x},T), \boldsymbol{u}'_\alpha(\mathbf{x},T)\right\rangle_{\dot{H}^1} = 2\left\langle \boldsymbol{u}^*_\alpha(\mathbf{x},0), \boldsymbol{u}'_\alpha(\mathbf{x},0)\right\rangle^2_{L^2}$$

- From the Riesz representation theorem

$$\Phi'_{\alpha,T}(\boldsymbol{\eta}:\boldsymbol{\eta}') = \left\langle \nabla\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha), \boldsymbol{\eta}'_\alpha\right\rangle_{G^\sigma}$$

- The gradient of the objective functional is

$$\nabla\Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha) = (1+|D|^2)^{-3} e^{-2\sigma|D|}\boldsymbol{u}^*_\alpha(x,0)$$

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- Tangent space to $\mathcal{M}_1$ at $\mathbf{v}$

$$\mathcal{T}_{\mathbf{v}}\mathcal{M}_1 = \left\{ z \in V : \langle \mathbf{v}, z \rangle_{\dot{H}^1} = 0 \right\}$$

- For any $z \in \mathcal{T}_{\mathbf{v}}\mathcal{M}_1$

$$\begin{aligned} \langle \mathbf{v}, z \rangle_{\dot{H}^1} &= \int_{\mathbb{T}^3} |D|\mathbf{v} \cdot |D|\mathbf{z}\, d\mathbf{x} \\ &= \int_{\mathbb{T}^3} (1 + |D|^2)^3 e^{2\sigma|D|} \left((1 + |D|^2)^{-3} e^{-2\sigma|D|} |D|^2 \mathbf{v} \right) \cdot \mathbf{z}\, d\mathbf{x} \\ &= \left\langle (1 + |D|^2)^{-3} e^{-2\sigma|D|} |D|^2 \mathbf{v}, \mathbf{z} \right\rangle_{G^\sigma} = 0 \end{aligned}$$

- Characterized using the unit 'normal' vector

$$\mathbf{n}_{\mathbf{v}} = \frac{(1 + |D|^2)^{-3} e^{-2\sigma|D|} |D|^2 \mathbf{v}}{\|(1 + |D|^2)^{-3} e^{-2\sigma|D|} |D|^2 \mathbf{v}\|_{G^\sigma}}$$

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- Projection operator $P_{\mathcal{T}_{\mathbf{v}}\mathcal{M}_1} : V \to \mathcal{T}_{\mathbf{v}}\mathcal{M}_1$

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- Retraction operator $R : V \to \mathcal{M}_1$

$$R(\mathbf{v}) = \frac{\sqrt{3}}{2} \frac{\mathbf{v}}{\|\mathbf{v}\|_{\dot{H}^1}}, \qquad \mathbf{v} \neq 0$$

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$$\beta_n = \frac{\left\langle P_{\mathcal{T}_n} \nabla \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha{}^{(n)}) - \Gamma_{\tau_{n-1}\mathbf{d}^{n-1}} P_{\mathcal{T}_{n-1}} \nabla \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha{}^{(n-1)})\right\rangle_{G^\sigma}}{\|P_{\mathcal{T}_{n-1}} \nabla \Phi_{\alpha,T}(\boldsymbol{\eta}_\alpha{}^{(n-1)})\|^2_{G^\sigma}}$$

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Numerical Methods

- Discretized in space using a standard pseudospectral Fourier method
- Derivatives are evaluated in the Fourier space
- Nonlinear products are computed in the physical space on an uniform grid with N points in each direction.
- Explicit fourth-order Runge-Kutta
- 2/3 dealiasing

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- Domain

$$\text{resolution} : 128^3, 256^3, 512^3, 1024^3$$
$$\alpha : 2^{-8}, 2^{-9}, 2^{-10}$$

- Time

$$dt : 2^{-5}$$
$$T : 3, 5$$

- Gevrey class

$$\sigma : 1E-5, 1E-4, 1E-3, 1E-2, 1E-1$$
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Results

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